

Differentiable functions – 2

Mean value theorems: how to produce a point ξ

1 сентября 2026 года

Theoretical minimum.

Theorem 1 (Rolle, Lagrange, Cauchy). If f is continuous on $[a, b]$, differentiable on (a, b) and $f(a) = f(b)$, then $f'(\xi) = 0$ for some ξ ; in general $f(b) - f(a) = f'(\xi)(b - a)$, and for a pair of functions $\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(\xi)}{g'(\xi)}$ provided $g' \neq 0$.

Theorem 2 (the auxiliary function). To prove the existence of ξ with $F(\xi, f(\xi), f'(\xi)) = 0$, construct a function vanishing at both endpoints and apply Rolle. The standard choices are $g(x)e^{\lambda x}$, $\frac{f(x)}{x}$, $f(x) - \ell(x)$ with a suitable affine ℓ , and $\varphi(x)f(x)$ with a solution φ of the associated homogeneous equation.

Fact 1. Two useful refinements: Darboux' theorem says the derivative has the intermediate value property, so f' cannot jump; and if f vanishes at $n+1$ points, then $f^{(n)}$ vanishes somewhere between the outermost of them.

1. (IMC 2019) Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be continuous with g differentiable, and suppose $(f(0) - g'(0))(g'(1) - f(1)) > 0$. Show that there is $\xi \in (0, 1)$ with $f(\xi) = g'(\xi)$.
2. (IMC 2023) Let V be the set of functions $f: [0, 1] \rightarrow \mathbb{R}$ differentiable on $(0, 1)$ with $f(0) = 0$ and $f(1) = 1$. Determine all $\alpha \in \mathbb{R}$ such that for every $f \in V$ there is $\xi \in (0, 1)$ with $f(\xi) + \alpha = f'(\xi)$.
3. (IMC 2013) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable with $f(0) = 0$. Prove that there is $\xi \in (-\frac{\pi}{2}, \frac{\pi}{2})$ with $f''(\xi) = f(\xi)(1 + 2 \tan^2 \xi)$.
4. (VJIMC 2015) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable. Prove that there is $\xi \in [0, 1]$ with $\frac{4}{\pi}(f(1) - f(0)) = (1 + \xi^2)f'(\xi)$.

5. (VJIMC 2016) Let $f: \mathbb{R} \rightarrow (0, \infty)$ be continuously differentiable. Prove that there is $\xi \in (0, 1)$ with $e^{f'(\xi)} f(0)^{f(\xi)} = f(1)^{f(\xi)}$.